

Interpretable and Uncertainty-Aware Crop Price Forecasting for Indian Markets using Temporal Fusion Transformers

Mini Project Report

Submitted in partial fulfillment of the requirements for the degree of
Master of Technology in Computer Science and Engineering

by

Devulapalli Tharun (252CS009)

under the guidance of

Prof. Shashidhar G. Koolagudi



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA,
SURATHKAL, MANGALORE – 575025

May, 2026

DECLARATION

I hereby declare that the **Mini Project** report entitled **Interpretable and Uncertainty-Aware Crop Price Forecasting for Indian Markets using Temporal Fusion Transformers** being submitted to the Department of Computer Science and Engineering, National Institute of Technology Karnataka, Surathkal, in fulfilment of the requirements of the M.Tech in CSE is a bona fide report of the work carried out by me. The material contained in this Report has not been submitted to any University or Institution for the award of any degree.

Devulapalli Tharun (252CS009)
Department of Computer Science and Engineering
NITK, Surathkal

Place: NITK, Surathkal.
Date: 04-05-2026

CERTIFICATE

This is to certify that the **Mini Project** report entitled **Interpretable and Uncertainty-Aware Crop Price Forecasting for Indian Markets using Temporal Fusion Transformers** submitted by **Devulapalli Tharun (252CS009)**, as the record of the work carried out, is accepted as the Mini Project work report submission in partial fulfillment for the award of the degree of Master of Technology in Computer Science and Engineering in the Department of Computer Science and Engineering, NITK, Surathkal.

Signature of Project Guide with date

Prof. Shashidhar G. Koolagudi,
Department of Computer Science and Engineering,
NITK, Surathkal

Abstract

Crop prices in India do not move smoothly. Onion prices in Delhi, Mumbai, and Hyderabad markets tripled between mid-2019 and early 2020, then fell sharply within months. Tomatoes followed the same pattern in mid-2023, crossing Rs. 200 per kg in several markets. These are not rare exceptions. They are a recurring pattern driven by monsoon variability, crop failures, export policy, and supply chain disruptions. For procurement planners and policy analysts, a model that reports one price number for next month without a range and without reasoning offers limited practical value.

This report presents an end to end forecasting system for monthly market-level crop prices across India. It combines World Food Programme (WFP) retail price records from 1994 to 2026 with NASA POWER (Prediction Of Worldwide Energy Resources) satellite weather data, and constructs a 26-feature panel covering onions, tomatoes, and rice across 53 markets. A Temporal Fusion Transformer (TFT) serves as the core probabilistic model, outputting multi-horizon quantile forecasts with interpretable variable weights and temporal attention. An XGBoost gradient boosted tree model runs in parallel as a point accuracy baseline.

The pipeline progresses through three modeling stages: a baseline TFT, an XGBoost-fused TFT variant, and per-crop Conformal Quantile Regression (CQR) calibration. The best result, TFT-XGBFusion-CQR, achieves a Mean Absolute Error (MAE) of 5.27 Rs per kg, a Mean Absolute Percentage Error (MAPE) of 11.4 percent, and 84.0 percent empirical coverage on the 2023 test window, with an average prediction band width of 11.99 Rs per kg. All outputs are delivered through a Streamlit dashboard that combines forecast trajectories, calibrated uncertainty bands, and interpretable model signals in one interface.

Temporal Fusion Transformer, XGBoost, Conformal Quantile Regression, Crop Price Forecasting, Indian Markets, Uncertainty Quantification, Interpretability, Streamlit

Contents

Abstract	ii
1 Introduction	1
1.1 Background	1
1.2 Motivation and Problem Context	1
1.3 Proposed Approach	1
1.4 Contributions	2
1.5 Objective	2
2 Literature Survey	4
2.1 Classical Statistical Approaches	4
2.2 Machine Learning Approaches	4
2.3 Deep Learning Approaches	4
2.4 Conformal Calibration	5
2.5 Gap in Existing Work	5
2.6 Proposed Direction	5
2.7 Literature Review Summary	5
3 Problem Statement	6
4 Dataset	7
4.1 Data Sources	7
4.2 Feature Engineering	7
4.3 Train-Test Split	8
4.4 Dataset Statistics	8
5 Methodology	9
5.1 Overall Pipeline	9
5.2 XGBoost Baseline	10
5.3 Temporal Fusion Transformer	10
5.3.1 Variable Selection Networks	10
5.3.2 LSTM Encoder-Decoder	10
5.3.3 Interpretable Multi-Head Attention and Quantile Heads	11
5.4 XGBoost-TFT Feature Fusion	11
5.5 Conformal Quantile Regression Calibration	12
5.6 Evaluation Metrics	12
6 Results and Analysis	13
6.1 Model Comparison	13
6.2 Ablation Study	13
6.3 Variable Importance Stability	14

6.4	Crop-wise Analysis	16
6.5	Comparison with Literature	18
6.6	Key Observations	18
7	Implementation and Dashboard	19
7.1	Technology Stack	19
7.2	Dashboard Views	19
7.3	Model Cascade Selection	21
8	Conclusion and Future Work	23
8.1	Conclusion	23
8.2	Future Work	23
	Bibliography	24

List of Figures

1.1	Conceptual pipeline for the proposed crop price forecasting system. .	2
5.1	Operational pipeline from data processing to dashboard deployment. .	9
5.2	TFT architecture: three input streams through VSN layers, LSTM encoder-decoder with static context injection, interpretable multi-head attention, and three quantile output heads.	10
6.1	Per-crop MAE across TFT variants (2023 test period). TFT-XGBFusion-CQR achieves the lowest MAE on all three crops.	14
6.2	Empirical coverage of the calibrated $q_{0.1}$ – $q_{0.9}$ interval (TFT-XGBFusion-CQR, 2023 test set). Dashed line shows the 80 percent nominal target. Bars statistically consistent with the target ($p > 0.05$) are shown in green.	14
6.3	Bootstrap stability of TFT encoder VSN importances (1000 resamples). Kendall $\tau = 0.942$ confirms stable feature ranking across market-crop series cohorts.	15
6.4	Comparison of encoder and decoder VSN feature importances. Weather related features and price memory variables dominate consistently across both pathways.	15
6.5	TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual onion prices on the held-out 2023 test set.	16
6.6	TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual tomato prices on the held-out 2023 test set. The band correctly widens during the mid-2023 price spike. .	17
6.7	TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual rice prices on the held-out 2023 test set. Rice shows the tightest intervals reflecting its stable price regime. .	17
7.1	Dashboard Price Forecast view: historical prices overlaid with the TFT-XGBFusion-CQR median forecast and calibrated uncertainty band. .	20
7.2	Dashboard Future Forecast view: month-by-month projection cards for the next six months with predicted median, calibrated interval, and risk classification.	20
7.3	Dashboard Explainability view: XGBoost gain-based feature importances (top), TFT encoder and decoder VSN weights (middle), and temporal attention heatmap (bottom).	21
7.4	Dashboard Present vs Predicted view: actual prices overlaid on model predictions for the 2023 held-out test period with the calibrated band as a shaded region.	21

List of Tables

1.1	Headline test metrics on the 2023 test window (792 observations) . . .	3
2.1	Representative related studies and methodological comparison	5
4.1	Data lineage from raw source to modeling dataset	7
4.2	Feature groups in the master dataset (26 columns total)	8
4.3	Temporal split for each model family	8
4.4	Crop coverage in the master dataset	8
5.1	TFT training configuration (consistent across all model families) . . .	11
6.1	Test metrics (2023, 792 rows) across all model families	13
6.2	Crop-wise test metrics for TFT-XGBFusion-CQR (2023 test set) . . .	16
6.3	Indicative comparison against selected published results	18
7.1	Core implementation stack	19

1 Introduction

1.1 Background

Crop prices in India swing unpredictably and at large magnitude. Onion prices in Delhi, Mumbai, and Hyderabad markets tripled between mid-2019 and early 2020 before falling sharply. Tomatoes followed the same pattern in mid-2023, with some markets crossing Rs. 200 per kg. These spikes are not rare tail events. They are a recurring feature driven by monsoon variability, crop failures, export restrictions, and logistical disruptions.

For households, sudden price increases shrink purchasing power within weeks. For food agencies and retail chains, they create procurement crises with little warning. A model that reports one number as next month’s price, without a range, without a risk flag, and without an explanation, is not useful in this environment. Forecasting at scale needs to answer three questions: what is the likely price, how confident should we be, and what is driving this estimate.

1.2 Motivation and Problem Context

Crop price forecasting in India is treated largely as a point accuracy problem in the published literature. Uncertainty quantification and interpretable reasoning are rarely integrated into deployed tools. The system built in this report treats both as first-class outputs. Three requirements follow naturally from the operational setting.

First, forecast accuracy: reduced absolute and percentage error over the test window. Second, uncertainty awareness: calibrated interval forecasts that are neither too wide to be useful nor too narrow to be reliable. Third, interpretability: attribution of predictions to specific features and time periods so that users can understand and trust the output.

1.3 Proposed Approach

The proposed system integrates four components: a Temporal Fusion Transformer (TFT) for multi-horizon probabilistic forecasting; an XGBoost model as a point accuracy baseline and as a feature source; Conformal Quantile Regression (CQR) calibration for reliable prediction intervals; and a Streamlit dashboard that presents all outputs in a unified decision-support interface. Figure 1.1 illustrates the full pipeline.

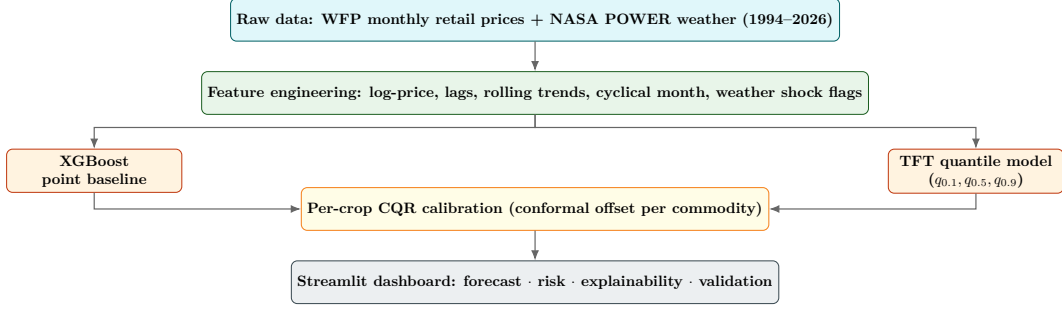


Figure 1.1: Conceptual pipeline for the proposed crop price forecasting system.

1.4 Contributions

The main contributions of this work are summarized as follows.

- **Reproducible multi-series pipeline:** A scripted data to model workflow for 53 markets and three crops, with all raw data, checkpoints, and calibration offsets version-controlled.
- **Calibrated probabilistic forecasting:** TFT-based quantile stack with per-crop CQR adjustment improving empirical coverage from 64.9 percent to 84.0 percent while simultaneously reducing prediction band width by 49 percent.
- **Leakage-controlled XGBoost fusion:** XGBoost predictions injected as known TFT covariates with a strict two-year temporal gap, achieving 50 percent MAE reduction and 60.8 percent MAPE reduction over TFT-Base.
- **Statistically validated interpretability:** Bootstrap analysis of Variable Selection Network (VSN) encoder weights (1000 resamples, Kendall $\tau = 0.942$) confirms stable feature attribution across market cohorts.
- **Unified decision interface:** Streamlit dashboard combining forecast trajectories, calibrated uncertainty, risk classification, VSN weights, and temporal attention in one reproducible workflow.

1.5 Objective

The primary objective is to develop a multi-series monthly price forecasting system for Indian crop markets that delivers accurate point estimates, calibrated prediction intervals, and interpretable feature attribution simultaneously. The system must be deployable through an interactive dashboard usable by procurement planners and analysts without requiring technical expertise in machine learning.

Table 1.1 summarizes the headline test metrics achieved.

Table 1.1: Headline test metrics on the 2023 test window (792 observations)

Model Variant	MAE (Rs/kg)	MAPE (%)	Coverage (%)	Band Width
XGBoost baseline	1.58	2.8	N/A	N/A
TFT-Base	10.53	29.1	64.9	23.62
TFT-EnsCQR	10.85	28.3	88.5	32.44
TFT-Retrain21-CQR	9.91	29.2	87.6	34.78
TFT-XGBFusion-CQR	5.27	11.4	84.0	11.99

2 Literature Survey

Agricultural crop price forecasting has a long history. Early statistical models gave way to machine learning regressors, which are now being overtaken by deep sequence models. Despite this progression, three gaps persist in the Indian crop price context: most studies optimise point accuracy but not calibrated uncertainty; interpretation of model outputs is rarely embedded in deployed tools; and direct comparison between probabilistic deep models and strong baselines under a unified interface is uncommon.

2.1 Classical Statistical Approaches

Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) models (Box and Jenkins, 2015) are interpretable and handle linear trend and seasonality, but fail under nonlinear shock dynamics common in Indian vegetable markets. Reported MAPE values above 30 percent are common for volatile crops such as onions and tomatoes. Seasonal decomposition methods face similar limitations when confronted with irregular supply shocks that break the underlying periodicity assumptions.

2.2 Machine Learning Approaches

Gradient boosted trees, especially XGBoost, improve nonlinear fitting and are effective when lag features are informative. XGBoost consistently outperforms ARIMA on monthly Indian crop price data, but produces no calibrated uncertainty intervals. Support vector regression and random forest models have also been applied, with comparable point accuracy but the same absence of probabilistic output.

Manogna et al. (2025) demonstrated RNN, LSTM, and GRU models alongside XGBoost on a large multi-crop, multi-market daily dataset, reporting onion MAPE of 14.59 percent for RNN and 21.42 percent for XGBoost. These are point forecasts only and provide no uncertainty quantification.

2.3 Deep Learning Approaches

LSTM and GRU models improve trend capture and allow multi-step forecasting. Transformer variants improve horizon-level representation learning. The Temporal Fusion Transformer (Lim et al., 2021) uniquely combines Variable Selection Networks (VSN), interpretable multi-head attention, and quantile output heads in a single architecture. Unlike prior approaches, TFT handles static metadata, known future covariates, and unknown historical inputs in separate processing streams, making it well suited to market-level panel data with exogenous weather variables.

Nayak et al. (2024) applied NBEATSX and XGBoost to weekly tomato, onion, and potato crop prices, reporting an onion NBEATSX sMAPE of 11.25 percent. This approach, while effective for weekly data, does not provide calibrated uncertainty or interpretable feature attribution.

2.4 Conformal Calibration

Romano et al. (2019) proposed Conformalized Quantile Regression (CQR) as a distribution-free post hoc method for constructing valid prediction intervals. It adjusts pre-trained quantile outputs using a calibration set without retraining the underlying model. The theoretical guarantee is marginal coverage: the true value falls within the adjusted interval at least $(1 - \alpha)$ fraction of the time under exchangeability. CQR is particularly useful because it is model-agnostic and computationally lightweight.

2.5 Gap in Existing Work

From the above discussion, four consistent gaps motivate the present work.

- Point accuracy is optimised but calibrated uncertainty is not evaluated.
- Strong baselines and probabilistic models are not compared under a unified interface.
- Coverage-oriented evaluation is under-reported in Indian crop price literature.
- Interpretability is rarely operationalised in deployed decision-support dashboards.

2.6 Proposed Direction

In contrast to prior work, this report addresses these gaps by combining probabilistic forecasting (TFT) with a strong point baseline (XGBoost), injecting XGBoost predictions as a feature to improve TFT accuracy, applying per-crop CQR calibration for reliable intervals, and operationalising interpretability through a deployed dashboard. This combination has not been demonstrated for Indian monthly crop price data at this scale.

2.7 Literature Review Summary

Table 2.1 summarizes the representative studies and compares them with the proposed approach.

Table 2.1: Representative related studies and methodological comparison

Study	Data	Model	Best reported	Limitations
Nayak et al. (2024)	TOP crops, weekly	NBEATSX, XGBoost	Onion 11.25%	sMAPE No uncertainty; no dashboard
Manogna et al. (2025)	23 commodities, daily	RNN/LSTM, XGBoost	Onion 14.59%	MAPE Point forecast only
Paul et al. (2022)	Brinjal, 17 markets	GRNN, RF, SVR	GRNN 3.68–17.86%	MAPE No uncertainty; one crop
TFT (2025)	tomato Tomato, 1 market	TFT variants	TFT 22.03%	sMAPE No fusion; no calibration
This work	3 crops, 53 markets	TFT + XGBoost + CQR	Onion 9.1%	MAPE Monthly retail only

3 Problem Statement

Crop price forecasting for Indian markets requires handling multiple sources of non-stationarity: seasonal cycles driven by monsoon patterns, structural shocks from export restrictions and supply chain disruptions, and cross-market correlations across geographically dispersed trading hubs. Traditional approaches that model each crop and market in isolation fail to leverage the shared temporal structure in the panel.

The specific challenge addressed in this work is: given monthly retail prices for a market-crop pair and a set of exogenous features, forecast the next H months with (a) a central forecast (the 50th percentile, or median, $q_{0.5}$), (b) a calibrated prediction interval defined by the 10th percentile ($q_{0.1}$, the low estimate) and the 90th percentile ($q_{0.9}$, the high estimate), and (c) interpretable feature importance weights. The 10th and 90th percentile bounds together form a central 80 percent prediction interval: in a well-calibrated model, the actual price should fall within this band 80 percent of the time. The test window covers 2023 onward, which includes the tomato price spike and post-COVID price normalisation, both of which are out-of-distribution events relative to the training period.

Formally, for a market-crop series with log-price y_t at month t :

$$\hat{y}_{t+h}^{(q)} = F_q(y_{t-L+1:t}, z_{t-L+1:t}, k_{t+1:t+H}, s), \quad h = 1, \dots, H, \quad q \in \{0.1, 0.5, 0.9\}.$$

Here $L = 24$ is the encoder look-back window, $H = 6$ is the prediction horizon, z are observed historical covariates, k are known future covariates (calendar and seasonal flags), and s is the static market-crop identity. Training uses pinball (quantile) loss:

$$\mathcal{L}_q(y, \hat{y}) = \begin{cases} q(y - \hat{y}), & y \geq \hat{y}, \\ (1 - q)(\hat{y} - y), & y < \hat{y}. \end{cases}$$

Empirical coverage over N test points is:

$$\text{Coverage} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i^{(0.1)} \leq y_i \leq \hat{y}_i^{(0.9)}].$$

The target is to minimise MAE and MAPE while maintaining empirical coverage above 80 percent at practically narrow interval widths. No prior work on Indian crop prices has simultaneously optimised all three objectives within a single deployed system.

4 Dataset

4.1 Data Sources

Two external sources are used. The World Food Programme (WFP) provides monthly retail crop price records for India under a CC BY license, covering January 1994 to February 2026 — over 30 years spanning multiple drought events, flood years, and the COVID-19 disruption. NASA POWER (Prediction Of Worldwide Energy Resources) provides satellite-derived monthly weather variables — temperature, rainfall, and humidity — aligned to each market’s GPS coordinates through its public API.

For each of the 53 markets in the modeling dataset, a market-to-GPS lookup was constructed. For each (market, year, month) combination, the NASA POWER monthly point API was queried for three variables: T2M (mean temperature at 2 m height), PRECTOTCORR (total monthly precipitation), and RH2M (mean relative humidity at 2 m height). Responses were cached locally to avoid repeated API calls.

Table 4.1: Data lineage from raw source to modeling dataset

Stage	Rows	Series/Markets	Notes
Raw WFP file	145,124	170 markets, 41 commodities	1994-01 to 2026-02
Filtered prices	18,415	123 series, 53 markets	Retail-only, series-length filter
Master dataset	16,939	123 series, 53 markets	After weather merge and lag warm-up removal

Three crops were chosen to represent distinct volatility regimes. Volatility is measured here by the Coefficient of Variation (CV), which is the ratio of the standard deviation to the mean expressed as a percentage. A high CV means prices fluctuate widely around the average. Onions have a CV of about 60 percent (highest volatility, monsoon driven supply shocks). Tomatoes have a CV of about 90 percent (highly event sensitive, with the 2023 price spike falling in the test window). Rice has a CV of about 20 percent (stable staple with government procurement support, which tests whether the model can also produce tight intervals for low-volatility series).

4.2 Feature Engineering

Six filtering steps produce the modeling dataset: remove aggregate rows; keep retail records only; retain target crops in kg units; normalise duplicate market labels; remove duplicate date-market keys; retain series with at least 60 monthly observations. Short gaps of three months or fewer are forward-filled; series with gaps longer than six months are dropped. A contiguous monthly integer index is assigned as $\text{time_idx} = 12(\text{year} - 1994) + (\text{month} - 1)$.

Table 4.2: Feature groups in the master dataset (26 columns total)

Group	Columns	Purpose
Identifiers	series_id, commodity, market, admin1, date, price	Series grouping and context
Temporal	time_idx, year, month, season, month_sin, month_cos, covid_lockdown	Calendar structure and known future covariates
Target and trend	log_price, price_lag_1m, price_lag_12m, rolling_3m, rolling_6m, yoy_change	Momentum, annual memory, trend per- sistence
Weather	temperature_mean, rainfall_monthly, humidity_mean	Exogenous environmental signals
Shock flags	rain_deficit, rain_excess, heat_stress, cold_stress	Binary indicators for adverse climate events

4.3 Train-Test Split

A strict chronological split is used throughout to prevent future data from leaking into the training set. If rows were split randomly, months from 2023 could appear in the training set, allowing the model to observe future prices during training and inflate test accuracy. Instead, the split follows calendar order as shown in Table 4.3.

Table 4.3: Temporal split for each model family

Family	Train	Validation	Test
TFT-Base	$\leq$ 2020-12-31 (13,227 rows)	2021–2022 (2,920 rows)	2023+ (792 rows)
TFT-XGBFusion / Retrain21	$\leq$ 2021-12-31 (14,686 rows)	2022 (1,461 rows)	2023+ (792 rows)

The XGBoost model used as the fusion covariate source is trained only on data up to 2019-12-31, providing a two-year gap relative to TFT training. All evaluation is performed on the 2023 held-out window (792 rows) across 123 market-crop series.

4.4 Dataset Statistics

Table 4.4: Crop coverage in the master dataset

Crop	Rows	Series	Markets	Role
Onions	5,003	38	38	Highest volatility; monsoon driven supply shocks
Tomatoes	5,245	45	45	Event sensitive; 2023 price spike in test window
Rice	6,691	40	40	Stable staple; tests interval tightness

5 Methodology

5.1 Overall Pipeline

The system is built around five pipeline stages: data build, model training, prediction and calibration, evaluation and explainability, and deployment. Figure 5.1 maps execution to these stages.

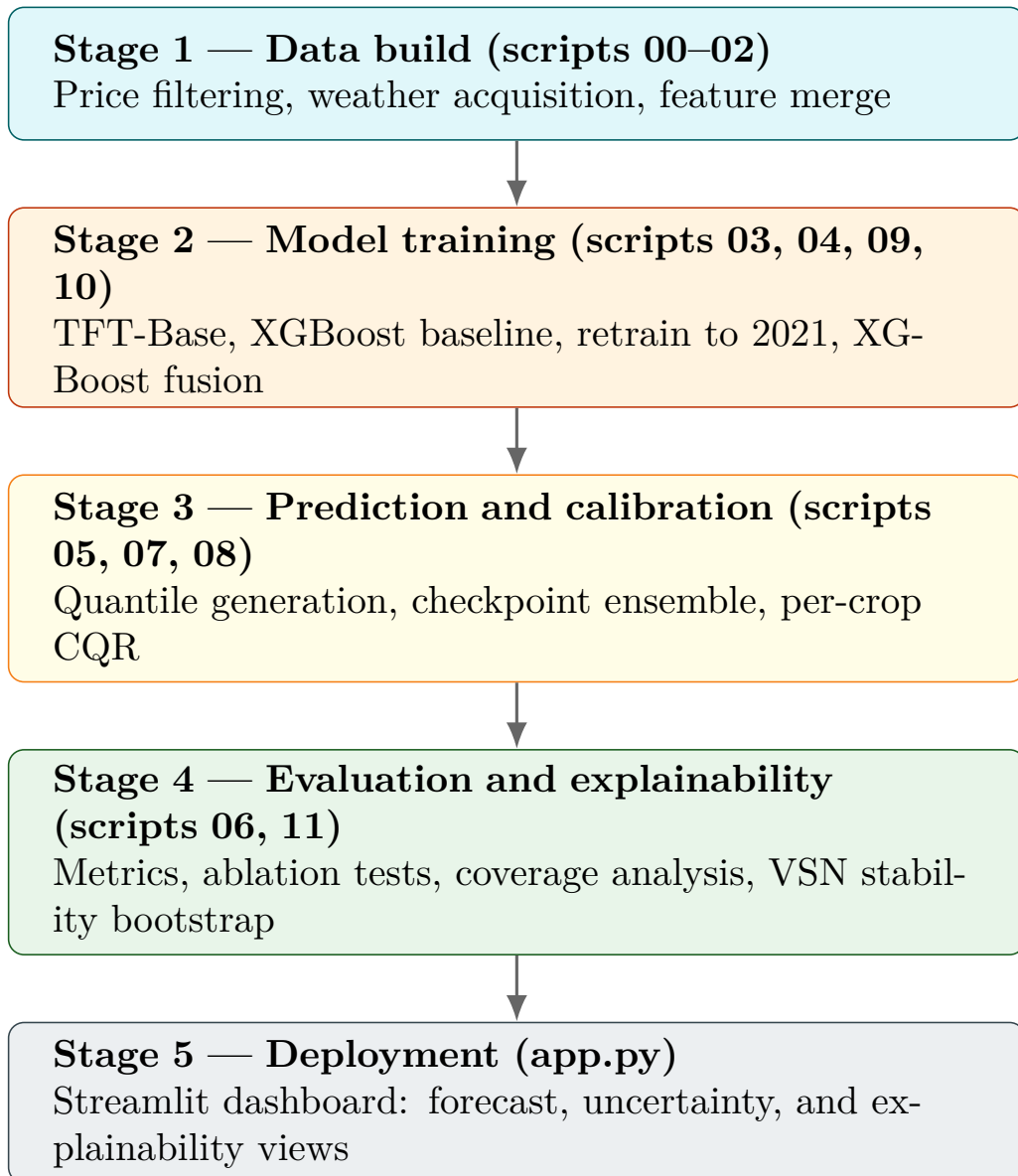


Figure 5.1: Operational pipeline from data processing to dashboard deployment.

5.2 XGBoost Baseline

An XGBoost gradient boosted tree regressor operates in log-price space:

$$\hat{y}_t = \sum_{m=1}^M \nu f_m(\mathbf{u}_t),$$

where each f_m is a regression tree, ν is the learning rate, and $\mathbf{u}_t$ includes the one-month lag, twelve-month lag, three-month and six-month rolling means, year on year change, and weather variables. The model is trained on data up to 2019-12-31 to maintain a two-year gap from the TFT training window. XGBoost achieves the lowest point forecast MAE (1.58 Rs per kg) because it directly minimises squared error over high-information lag features. It provides no calibrated uncertainty interval.

5.3 Temporal Fusion Transformer

The Temporal Fusion Transformer (Lim et al., 2021) handles multi-horizon quantile forecasting with built-in interpretability. Figure 5.2 shows its structure: three input streams through Variable Selection Networks (VSN), an LSTM encoder-decoder with static context injection, interpretable multi-head attention, and quantile output heads.

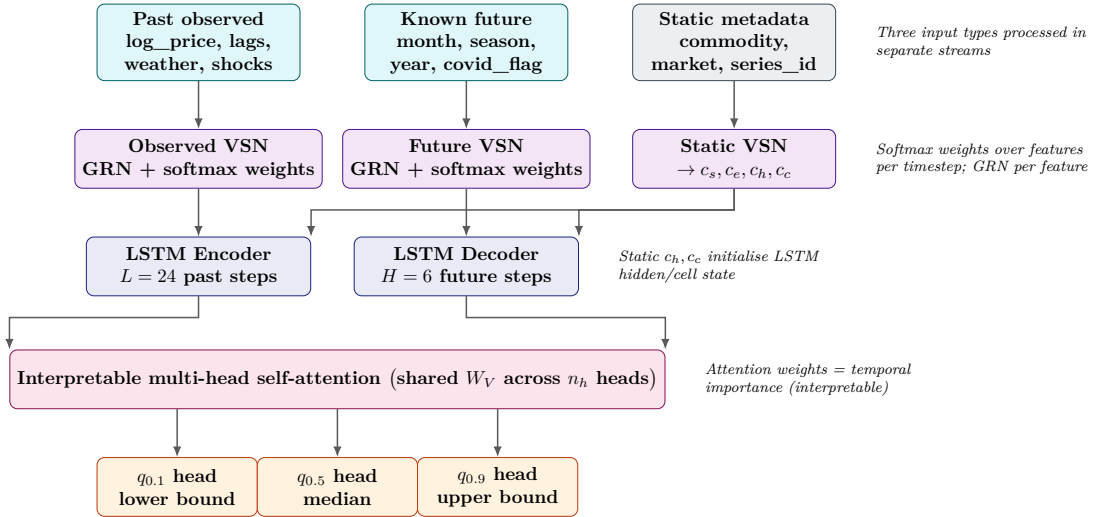


Figure 5.2: TFT architecture: three input streams through VSN layers, LSTM encoder-decoder with static context injection, interpretable multi-head attention, and three quantile output heads.

5.3.1 Variable Selection Networks

Variable Selection Networks (VSN) learn per-timestep softmax weights over all features via Gated Residual Networks (GRN). For each timestep, a GRN is applied to each input feature independently, and a second GRN produces the softmax weights over all features. The output is a weighted combination of transformed features. These weights are the primary interpretability output of TFT: they indicate which features the model relies on most at each point in the input sequence.

5.3.2 LSTM Encoder-Decoder

The LSTM encoder reads 24 past observations and produces a sequence of hidden states. The LSTM decoder processes 6 future steps. Static embeddings derived from

the Static VSN initialise the LSTM hidden and cell states, allowing the model to condition forecasts on the identity of the market and crop. A gating mechanism controls how much of each hidden state passes to the attention layer.

5.3.3 Interpretable Multi-Head Attention and Quantile Heads

The attention layer uses a shared value matrix W_V across all heads. This makes the aggregated attention weights interpretable as position-level temporal importance: high attention on month $t - k$ means the model found historical context k steps ago informative for the current forecast. Three quantile output heads independently produce $q_{0.1}$, $q_{0.5}$, and $q_{0.9}$ for each future horizon, trained jointly with pinball loss.

The 10th percentile head ($q_{0.1}$) gives the low price estimate, the 50th percentile head ($q_{0.5}$) gives the median or central prediction, and the 90th percentile head ($q_{0.9}$) gives the high price estimate. Together these form a central 80 percent prediction interval.

Table 5.1: TFT training configuration (consistent across all model families)

Parameter	Value	Rationale
hidden_size	32	Compact for monthly panel; prevents overfitting
attention_head_size	2	Balance between diversity and stability
lstm_layers	1	Limits overfitting in small to medium panel
dropout	0.2	Cross-series regularisation
quantiles	0.1, 0.5, 0.9	Direct probabilistic forecasting
max_encoder_length	24 months	Annual cycle plus trend memory
max_prediction_length	6 months	Short to medium planning horizon
learning_rate	0.03	Stable convergence under Ranger optimizer

5.4 XGBoost-TFT Feature Fusion

In TFT-XGBFusion, the log-price prediction from the pre-2020 XGBoost model is injected as a known covariate (`xgb_log_pred`) at every future timestep. This design exploits XGBoost’s local lag accuracy as a structured prior for TFT while preserving a strict two-year gap: the XGBoost model is trained on data up to 2019-12-31, and TFT training uses data up to 2021-12-31.

This approach is fundamentally different from model stacking (Wolpert, 1992), in which two models predict independently and their outputs are blended afterward. Here TFT receives the XGBoost prediction as a feature and decides through its VSN how much weight to assign to it at each timestep. For a stable rice market the XGBoost signal is likely accurate and TFT can assign high weight to it. During a tomato spike, if XGBoost underestimates because tree models do not extrapolate to previously unseen price levels, TFT can downweight that signal and rely on its own sequential reasoning.

The result is a 50 percent MAE reduction (from 10.53 Rs per kg to 5.27 Rs per kg) and a 60.8 percent MAPE reduction (from 29.1 percent to 11.4 percent) over TFT-Base, while also reducing the calibrated interval width by 49 percent.

5.5 Conformal Quantile Regression Calibration

TFT-Base achieves only 64.9 percent coverage on the test set, well below expectation for a $q_{0.1}$ – $q_{0.9}$ interval. CQR (Romano et al., 2019) corrects this without retraining. For each calibration sample i :

$$E_i = \max(q_{0.1,i} - y_i, y_i - q_{0.9,i}).$$

For target miscoverage $\alpha = 0.10$, the per-crop offset is:

$$\hat{Q}_\alpha = \text{Quantile}_{1-\alpha}(\{E_i\}_{i=1}^n).$$

Calibrated bounds are then:

$$\tilde{q}_{0.1} = q_{0.1} - \hat{Q}_\alpha, \quad \tilde{q}_{0.9} = q_{0.9} + \hat{Q}_\alpha.$$

The median $q_{0.5}$ is unchanged. Calibration offsets are computed separately per crop, allowing the system to account for the different tail behaviour of onions, tomatoes, and rice. A global offset would over-widen the rice interval and under-widen the tomato interval.

5.6 Evaluation Metrics

Point accuracy is measured by MAE (Rs per kg), RMSE (Rs per kg), and MAPE (percent) in original price scale after exponentiating log-space predictions. Interval quality is measured by empirical coverage and mean band width. Statistical significance of improvements is assessed using paired t -tests and Wilcoxon signed-rank tests, and effect size is reported as Cohen’s d . Interpretability stability is assessed through 1000-resample bootstrap of VSN encoder weights with mean pairwise Kendall τ .

6 Results and Analysis

6.1 Model Comparison

Table 6.1 presents the full comparison of all model families on the 2023 test window. Two patterns stand out. First, XGBoost dominates on point accuracy, reaching an MAE of 1.58 Rs per kg by directly minimising squared error over high-information lag features. Second, TFT-Base’s 64.9 percent coverage confirms that raw quantile training is insufficient for reliable uncertainty bounds. CQR calibration is essential, not optional.

Table 6.1: Test metrics (2023, 792 rows) across all model families

Model variant	MAE (Rs/kg)	MAPE (%)	Coverage (%)	Band width
XGBoost baseline	1.58	2.8	N/A	N/A
TFT-Base	10.53	29.1	64.9	23.62
TFT-EnsCQR	10.85	28.3	88.5	32.44
TFT-Retrain21-CQR	9.91	29.2	87.6	34.78
TFT-XGBFusion-CQR	5.27	11.4	84.0	11.99

6.2 Ablation Study

Relative to TFT-Base, TFT-XGBFusion-CQR achieves 60.8 percent MAPE reduction (29.1 to 11.4 percent) and 50.0 percent MAE reduction (10.53 Rs per kg to 5.27 Rs per kg). Coverage improves by 19.1 percentage points while band width decreases by 49.2 percent. Critically, TFT-EnsCQR reaches 88.5 percent coverage but at a width of 32.44 Rs per kg, nearly three times wider than TFT-XGBFusion-CQR at 11.99 Rs per kg. Wider coverage alone is not a better outcome; the fusion model achieves a superior accuracy-calibration balance.

Paired statistical tests confirm the improvement over TFT-Base: $t = -22.34$, $p = 4.6 \times 10^{-86}$, Cohen’s $d = -0.79$ (large effect), Wilcoxon $p = 1.2 \times 10^{-95}$. The step from TFT-Base to TFT-Retrain21-CQR (adding only extended cutoff) shows a smaller but significant contribution: $t = -4.15$, $p = 3.6 \times 10^{-5}$, Cohen’s $d = -0.15$.

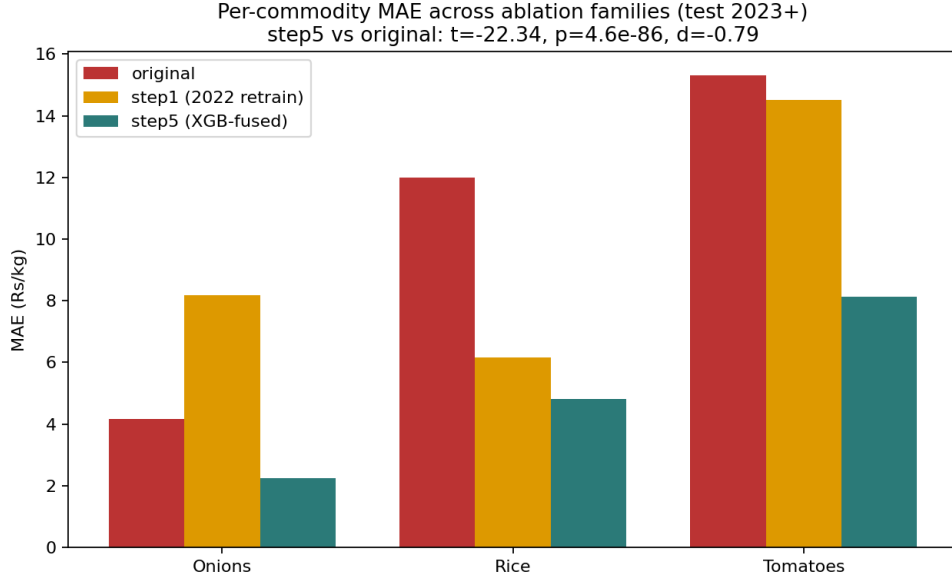


Figure 6.1: Per-crop MAE across TFT variants (2023 test period). TFT-XGBFusion-CQR achieves the lowest MAE on all three crops.

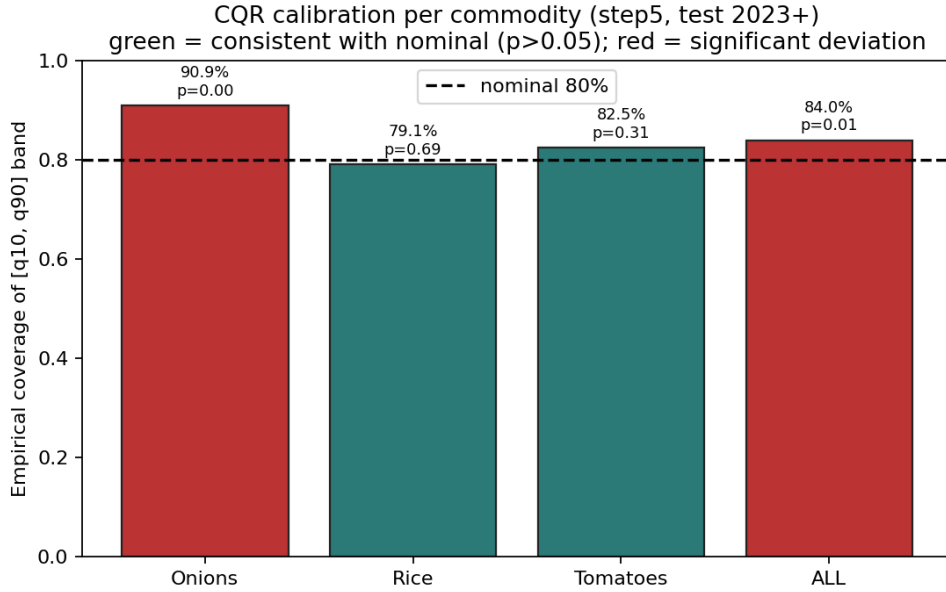


Figure 6.2: Empirical coverage of the calibrated $q_{0.1}-q_{0.9}$ interval (TFT-XGBFusion-CQR, 2023 test set). Dashed line shows the 80 percent nominal target. Bars statistically consistent with the target ($p > 0.05$) are shown in green.

6.3 Variable Importance Stability

Bootstrap analysis (1000 series-level resamples) of VSN encoder weights yields mean pairwise Kendall $\tau = 0.942$ with a 95 percent confidence interval of $[0.895, 0.990]$. This indicates that the feature ranking is highly stable across different subsets of the market-crop panel, which is a necessary condition for trusting VSN weights as interpretable signals.

Top encoder features by mean importance: temperature_mean (0.1432), rain_excess (0.1427), rain_deficit (0.1402), humidity_mean (0.0664), year (0.0600), covid_lockdown (0.0587), season (0.0419), log_price (0.0405), rolling_6m (0.0382), price_lag_12m (0.0381). Weather variables dominate the top three, consistent with monsoon driven supply shocks being the primary driver of price volatility in Indian vegetable markets.

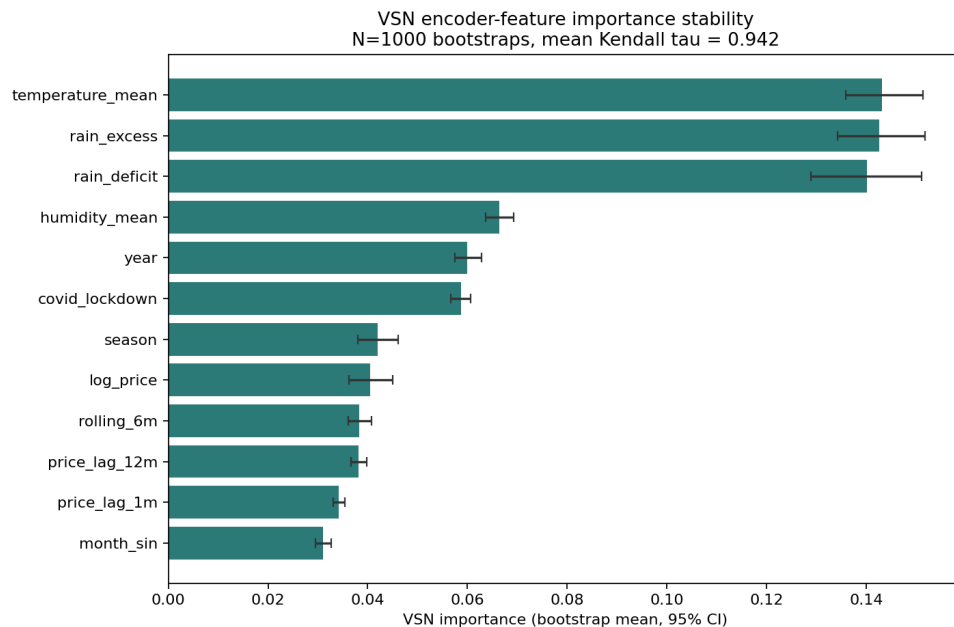


Figure 6.3: Bootstrap stability of TFT encoder VSN importances (1000 resamples). Kendall $\tau = 0.942$ confirms stable feature ranking across market-crop series cohorts.

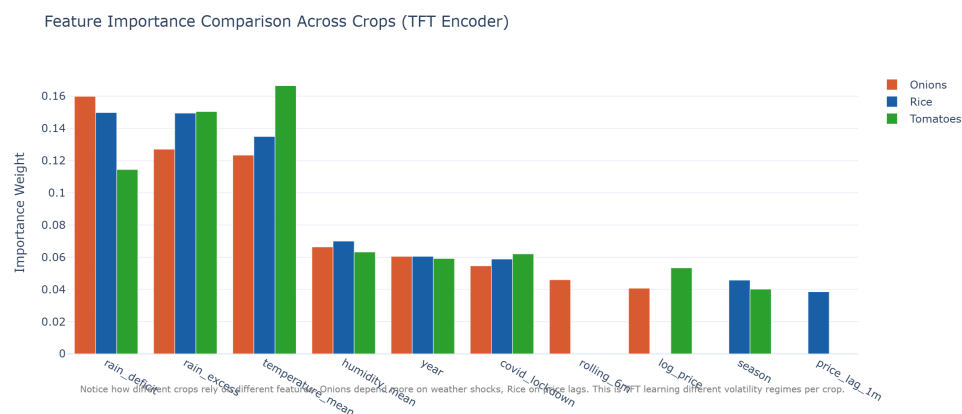


Figure 6.4: Comparison of encoder and decoder VSN feature importances. Weather related features and price memory variables dominate consistently across both pathways.

6.4 Crop-wise Analysis

Table 6.2: Crop-wise test metrics for TFT-XGBFusion-CQR (2023 test set)

Crop	MAE (Rs/kg)	MAPE (%)	Coverage (%)	Observations
Onions	2.25	9.1	90.9	Best accuracy; conservative interval
Tomatoes	8.13	13.0	82.5	Hardest series; 2023 spike in test window
Rice	4.81	11.7	79.1	Stable regime; narrowest intervals
Overall	5.27	11.4	84.0	n=792 observations

Figures 6.5–6.7 show the calibrated TFT-XGBFusion-CQR prediction range overlaid on actual prices for each crop in the 2023 test period. The shaded band represents the central 80 percent prediction interval ($q_{0.1}^{\text{cal}}$ to $q_{0.9}^{\text{cal}}$) and the solid line is the median prediction. The onion and rice forecasts track actual prices closely. The tomato forecast correctly produces a wider band during the mid-2023 price surge, reflecting higher prediction uncertainty for that crop.

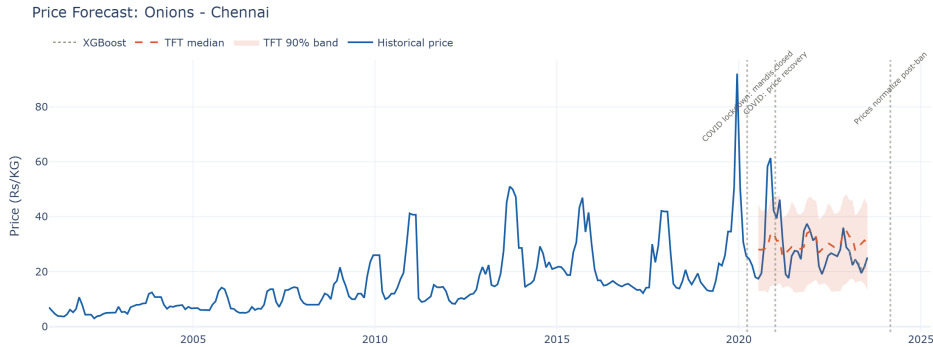


Figure 6.5: TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual onion prices on the held-out 2023 test set.

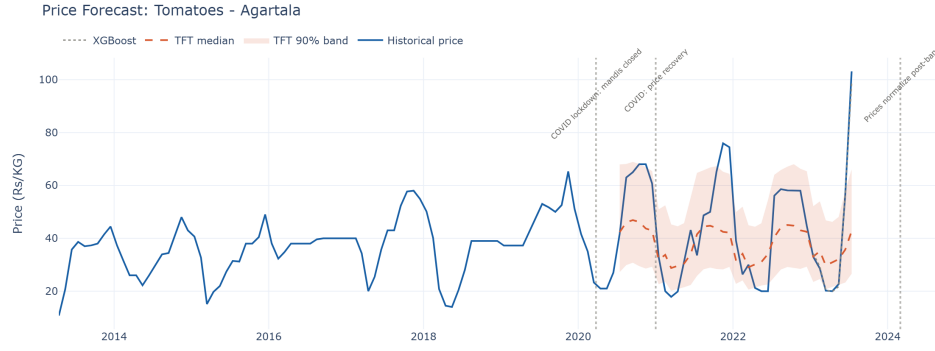


Figure 6.6: TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual tomato prices on the held-out 2023 test set. The band correctly widens during the mid-2023 price spike.



Figure 6.7: TFT-XGBFusion-CQR calibrated prediction range (shaded) and median forecast (line) versus actual rice prices on the held-out 2023 test set. Rice shows the tightest intervals reflecting its stable price regime.

Tomato is the most challenging crop. Its 2023 price spike, driven by supply disruptions not visible in 24 months of price and weather history, compresses into monthly averages. The 82.5 percent coverage shows CQR absorbing most spike risk through wider intervals. Onions achieve 90.9 percent coverage because monsoon driven shocks align well with the weather features. Rice's 79.1 percent coverage marginally falls short of the 80 percent target, indicating that the CQR offset slightly under-covers the rice distribution tail.

6.5 Comparison with Literature

Table 6.3: Indicative comparison against selected published results

Study	Target	Reported metric	Note
Nayak et al. (2024)	Onion	NBEATsX sMAPE 11.25%	Weekly; no uncertainty
Manogna et al. (2025)	Onion	RNN MAPE 14.59%	Daily multi-market
Paul et al. (2022)	Brinjal	GRNN MAPE 3.68–17.86%	Market range; one crop
This work	Onion	MAPE 9.1%	Monthly; calibrated bands and interpretability

These values are not directly comparable because frequency and dataset scope differ across studies. The onion MAPE of 9.1 percent on monthly data is competitive with weekly-data deep models in recent literature, while additionally providing calibrated uncertainty bands and statistically validated interpretable feature rankings.

6.6 Key Observations

- XGBoost provides the strongest point accuracy and should not be dismissed as a baseline; its 1.58 Rs per kg MAE reflects how informative monthly lag features are.
- CQR calibration is essential: TFT-Base’s 64.9 percent raw coverage is far below the 80 percent expectation and makes the model unsuitable as a risk-facing decision tool.
- The XGBoost fusion design is the most effective single intervention: injecting a clean XGBoost signal as a known covariate simultaneously improves point accuracy and narrows calibrated intervals.
- VSN feature rankings are stable (Kendall $\tau = 0.942$) and can be treated as model-internal attribution signals, though they do not constitute causal proof of any relationship between weather and price.
- Tomatoes remain the hardest crop across all metrics, representing the strongest case for incorporating supply-event covariates in future work.

7 Implementation and Dashboard

7.1 Technology Stack

Table 7.1: Core implementation stack

Layer	Technology	Purpose
Data engineering	pandas, numpy	Filtering, joins, feature generation
Weather ingestion	requests + NASA POWER API	Monthly exogenous covariate acquisition
Deep forecasting	PyTorch, PyTorch Forecasting, Lightning	TFT training and inference
Baseline modeling	XGBoost + joblib	Point forecast and artifact serialization
Evaluation and plots	Plotly, Kaleido	Metrics and visualisation generation
Application layer	Streamlit	Interactive forecast and explainability dashboard

Training was run on an NVIDIA GPU (CUDA 12.8). Each TFT family takes 30 to 90 minutes on GPU and 4 to 8 hours on CPU. Dashboard inference runs on CPU in real time with artifact loading times under two seconds.

7.2 Dashboard Views

The Streamlit dashboard exposes four views accessible through a sidebar crop-market selector.

- **Price Forecast:** Historical price series overlaid with TFT-XGBFusion-CQR median and the calibrated $q_{0.1}$ – $q_{0.9}$ band. Vertical markers flag months where the relative price change exceeds 25 percent. An optional XGBoost overlay enables direct comparison of probabilistic and point forecasts on the same chart.
- **Future Forecast:** Month-by-month projection cards for the next six months, each showing the predicted median, the calibrated interval, and a risk classification (Stable, Moderate, or High) derived from the relative interval width $r_t = (q_{0.9,t} - q_{0.1,t}) / q_{0.5,t}$.
- **Explainability:** XGBoost feature importances, TFT encoder and decoder VSN weight bar charts, and a temporal attention heatmap showing which past months the model attended to most. Weather variables consistently dominate encoder importance, consistent with the bootstrap result.

- **Present vs Predicted:** Test-period chart comparing actual prices to TFT predictions, alongside a per-metric summary table (MAE, RMSE, MAPE, coverage, band width). Enables operational validation without re-running the evaluation pipeline.

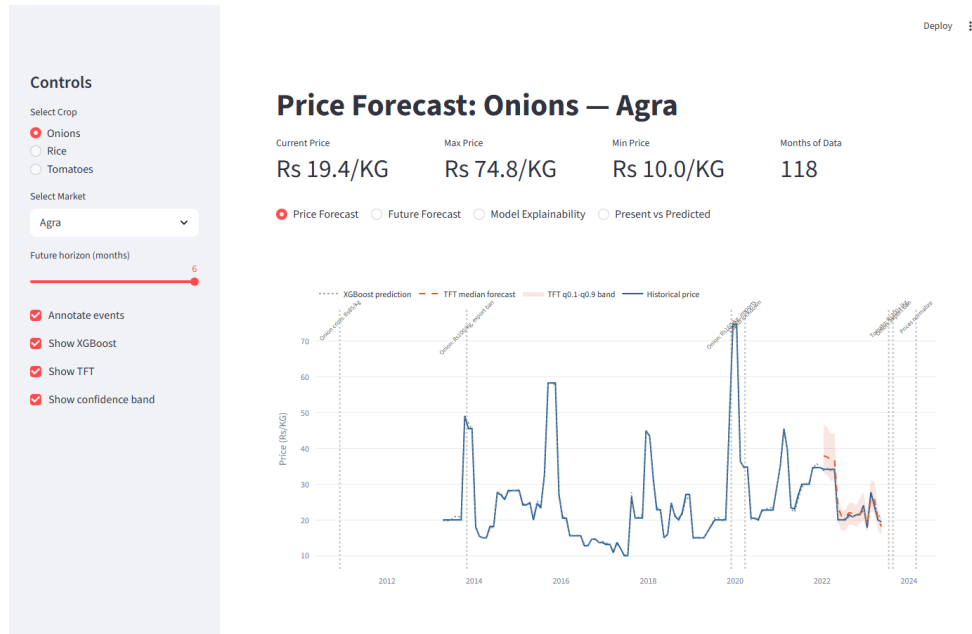


Figure 7.1: Dashboard Price Forecast view: historical prices overlaid with the TFT-XGBFusion-CQR median forecast and calibrated uncertainty band.

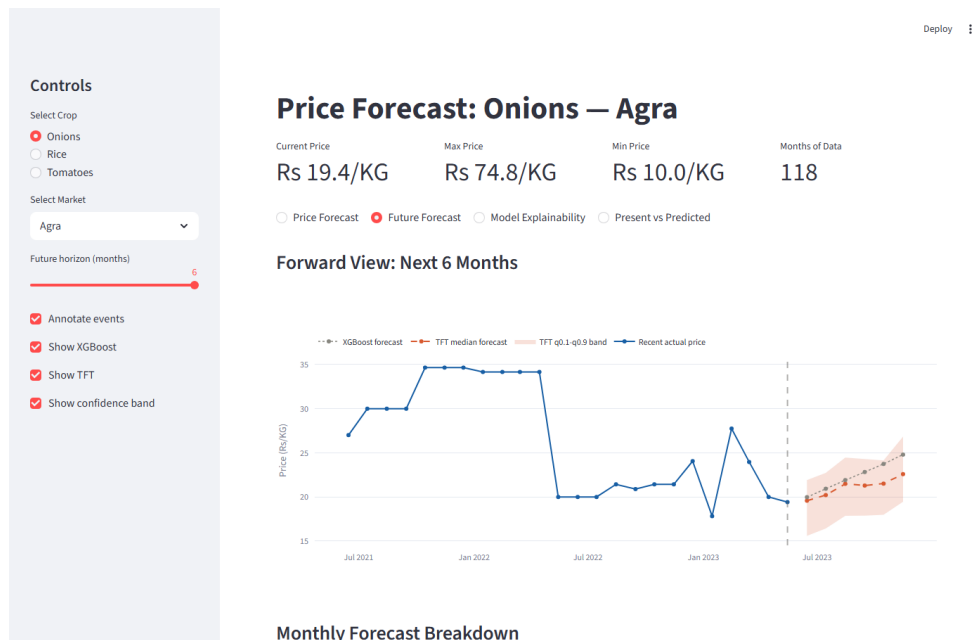


Figure 7.2: Dashboard Future Forecast view: month-by-month projection cards for the next six months with predicted median, calibrated interval, and risk classification.

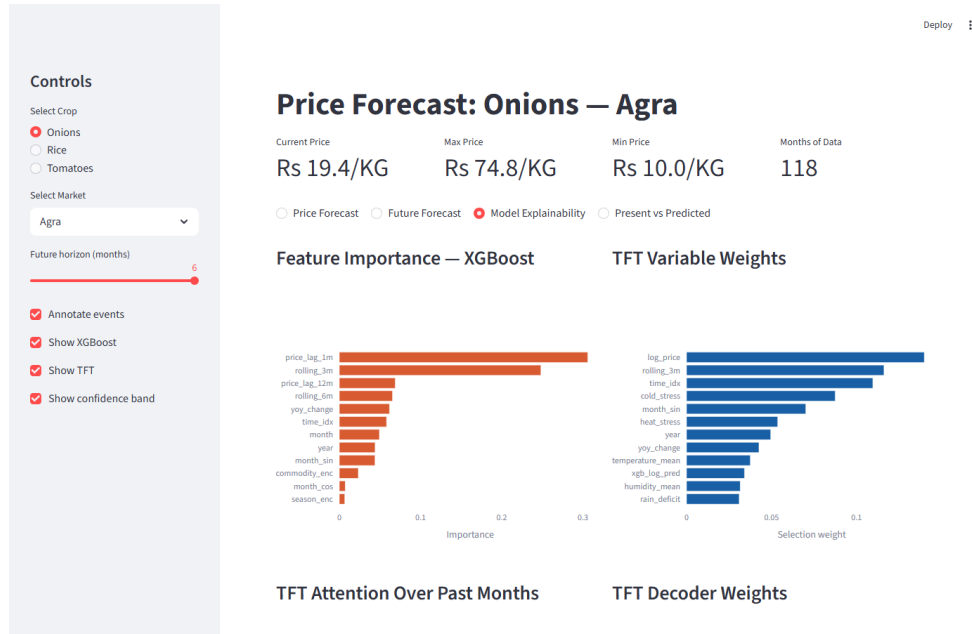


Figure 7.3: Dashboard Explainability view: XGBoost gain-based feature importances (top), TFT encoder and decoder VSN weights (middle), and temporal attention heatmap (bottom).

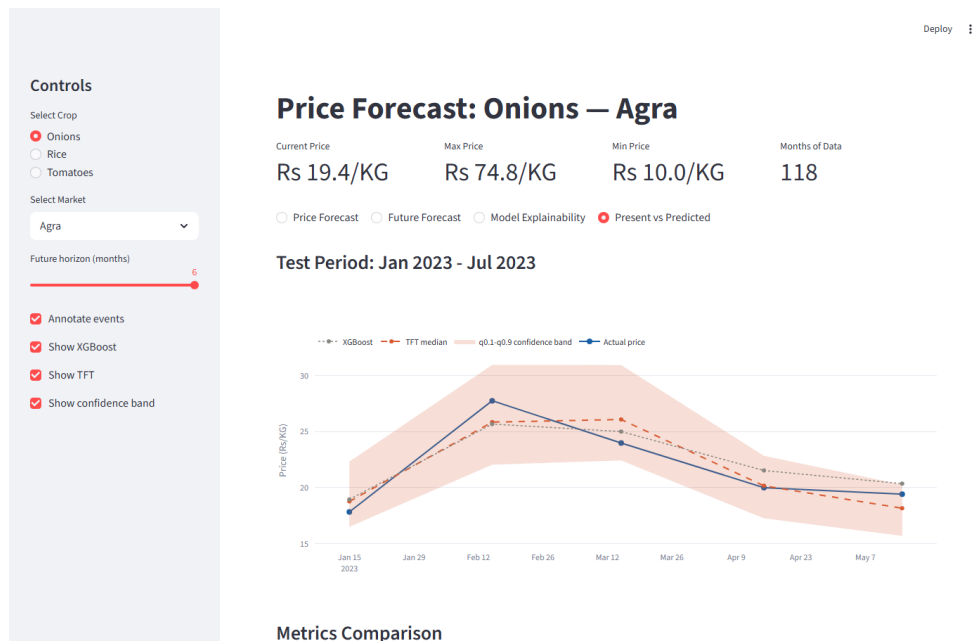


Figure 7.4: Dashboard Present vs Predicted view: actual prices overlaid on model predictions for the 2023 held-out test period with the calibrated band as a shaded region.

7.3 Model Cascade Selection

At startup, the application checks which TFT artifacts are available on disk. If the fused checkpoint and dataset are present, it selects TFT-XGBFusion-CQR as the active probabilistic model. If not present, it falls back to TFT-Retrain21-CQR, and then to TFT-Base. CQR calibration offsets are loaded from the corresponding `conformal_offsets_*.json` file. This cascading selection means the dashboard

always uses the best available model without requiring manual configuration.

8 Conclusion and Future Work

8.1 Conclusion

This mini project built an interpretable, uncertainty-aware crop price forecasting system for Indian markets. The system covers three crops (onions, tomatoes, rice) across 53 markets in a monthly panel spanning 1994 to 2026, and delivers probabilistic forecasts with calibrated uncertainty bands and statistically validated interpretable feature attribution.

Among all model families evaluated, TFT-XGBFusion-CQR provides the strongest practical balance: an MAE of 5.27 Rs per kg, a MAPE of 11.4 percent, and 84.0 percent empirical coverage on the 2023 test window. Paired statistical tests ($p = 4.6 \times 10^{-86}$, Cohen’s $d = -0.79$) and bootstrap VSN stability analysis (Kendall $\tau = 0.942$) confirm that the improvements are consistent and not artifacts of random variation.

XGBoost remains the strongest pure point model but does not provide calibrated uncertainty or interpretable temporal reasoning. The deployed Streamlit dashboard operationalises the broader objective: actionable crop price forecasting with quantified risk and inspectable model reasoning, in one reproducible workflow.

8.2 Future Work

1. **News and event integration.** The dashboard already fetches commodity news for qualitative context. The direct next step is building a pipeline to ingest news headlines, apply commodity-specific sentiment scoring, and inject the resulting signal as an exogenous TFT covariate. This is the most targeted fix for tomato’s abrupt 2023-type spikes.
2. **Richer supply-side covariates.** Adding market arrivals from AG-MARKNET, export ban episodes, cold storage capacity, and fuel price index would address the largest identified gap in the current feature set. These policy and logistics variables drove both the 2019 onion and 2023 tomato episodes.
3. **Rolling conformal recalibration.** Static CQR offsets are computed from a fixed historical validation window. A rolling recalibration that updates offsets as new data arrives would adapt to regime shifts faster, particularly for tomato markets.
4. **Higher-frequency expansion.** Weekly or daily price data from AG-MARKNET would multiply the training set size, improve short-term spike detection, and allow shorter recalibration windows. The current monthly framework is a practical starting point, not a fundamental constraint.
5. **Broader crop and market coverage.** Expanding to pulses, spices, and edible oils, and extending to more state-level mandi data, would increase utility for national agricultural policy monitoring. External validation on post-2024 data is also needed before production deployment.

Bibliography

1. Lim, B., Arik, S. O., Loeff, N., and Pfister, T. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting*, 37(4), 1748–1764.
2. Romano, Y., Patterson, E., and Candes, E. (2019). Conformalized Quantile Regression. *Advances in Neural Information Processing Systems (NeurIPS)*.
3. Nayak, G. H. H., Alam, M. W., Singh, K. N., et al. (2024). Exogenous Variable Driven Deep Learning Models for Improved Price Forecasting of TOP Crops in India. *Scientific Reports*.
4. Manogna, R. L., Dharmaji, V., and Sarang, S. (2025). Enhancing Agricultural Commodity Price Forecasting with Deep Learning. *Scientific Reports*.
5. Paul, R. K., Yeasin, M., et al. (2022). Machine Learning Techniques for Forecasting Agricultural Prices: A Case of Brinjal in Odisha, India. *PLOS ONE*.
6. Box, G. E. P. and Jenkins, G. M. (2015). Time Series Analysis: Forecasting and Control, 5th ed. John Wiley and Sons, Hoboken, NJ.
7. Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.
8. World Food Programme (2026 access). WFP Food Prices for India Dataset. <https://data.humdata.org/dataset/wfp-food-prices-for-india>.
9. NASA POWER Project (2026 access). Monthly Point API. <https://power.larc.nasa.gov>.
10. PyTorch Forecasting Documentation (2026 access). <https://pytorch-forecasting.readthedocs.io>.
11. PyTorch Lightning Documentation (2026 access). <https://lightning.ai/docs/pytorch>.
12. Streamlit Documentation (2026 access). <https://docs.streamlit.io>.